Complete Work Plan: Experiments & Implementation

Learning to Recommend in Unknown Games — NeurIPS 2026

April 2026 — v2 (post literature survey)

1 Current Status

1.1 What is implemented and tested

- **[Done]** Normal-form game environment with BR, QR, and γ-BR feedback
- **[Done]** CE solver (Gurobi, sparse, scales to M=32,768)
- **[Done]** Separation oracle (Algorithm 4, 48 tests passing)
- **[Done]** Cutting-plane algorithm (ACCPM with Woodbury, scales to N=24,576)
- **[Done]** Player swap regret tracking
- **[Done]** MW baseline (no moderator)
- **[Done]** Preliminary plots (7 experiments, all running)

1.2 What is missing (from literature survey)

- **[TODO]** Canonical benchmark games (Prisoner's Dilemma, Shapley, Braess)
- **[TODO]** ε-CE gap / NashConv metric
- **[TODO]** Baselines: random recommendation, oracle, regret matching
- **[TODO]** Log-log regret plots
- **[TODO]** Compact pipeline for congestion games
- **[TODO]** Sioux Falls or real-network experiment

2 Work Plan: Phase 1 — Benchmark Games and Metrics

Estimated time: 2–3 hours coding + 1 hour running.

2.1 Task 1.1: Add canonical benchmark games

Implement named game constructors in `game.py`:

1. **Prisoner's Dilemma** (2 players, 2 actions):

$$u_1 = \begin{pmatrix} -1 & -3 \\ 0 & -2 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -1 & 0 \\ -3 & -2 \end{pmatrix}$$

2. **Shapley Game** (2 players, 3 actions): the classic game where best-response dynamics cycle but CE exists.

$$u_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

3. **Matching Pennies** (2 players, 2 actions): zero-sum.

4. **Chicken / Hawk-Dove** (2 players, 2 actions): multiple NE, non-trivial CE.
5. **Theorem 5 counterexample** (2 players, 4×2): already in tests, expose as named constructor.

2.2 Task 1.2: Add ε-CE gap metric (NashConv)

The ε-CE gap is the maximum incentive to deviate from the recommendation:

$$\varepsilon\text{-CE}(x) \;=\; \max_{i \in [n]} \; \max_{a_i, a_i' \in A_i} \; \phi_i(a_i, a_i', x)$$

This is the standard metric in the no-regret $\to$ CE literature. Implement as a method on `NormalFormGame` and track it per-round in `CuttingPlaneAlgorithm`.

Add to `CuttingPlaneResult`: field `ce_gap: NDArray` of shape $(T,)$.

2.3 Task 1.3: Add baselines

Add to `baselines.py`:

1. **Random recommendation:** $x(t) = \text{Uniform}(\Delta(A))$ every round. No learning. Lower bound.
2. **Oracle:** $x(t) \in \text{CE}(w^\star)$ (true game known). Zero compliance regret after first round. Upper bound.
3. **Regret matching** (Hart–Mas-Colell): each player i maintains regret vector $R_i(a_i, a_i')$ and plays proportional to positive regrets. Standard CE convergence baseline. Converges at $\mathcal{O}(\sqrt{T})$.
4. **EXP3 per player:** bandit baseline, each player learns independently. Already implemented (MW baseline). Rename for clarity.

3 Work Plan: Phase 2 — Main Paper Experiments (4 Figures)

Estimated time: 3–4 hours running (parallelizable).

3.1 Figure 1: Compliance Regret vs T (Theorem 8)

Goal: Show $R(T) = \mathcal{O}(nM \log T)$.

- **Panel A:** Regret vs t on log-log scale. Curves for Prisoner's Dilemma (nM=8), Shapley (nM=18), $(2,2,2)$ (nM=24), $(3,3,3)$ (nM=81), $(2,)^{\times 5}$ (nM=160).
- **Panel B:** $R(t)/\log(t)$ vs t. Should plateau at constant $\propto nM$.
- **Reference lines:** $c \cdot nM \cdot \log(t)$ fitted per curve.
- Games: 5 games (3 canonical + 2 random). $T = 5{,}000$. Seeds: 30. Feedback: BR (γ=0).
- **Baselines on same plot:** Random recommendation (linear regret), Oracle (zero).

3.2 Figure 2: Learnability Separation — QR vs BR (Theorems 4 & 5)

Goal: QR converges to equivalence class, BR does not.

- **Panel A (QR):** Distance to equivalence class $d(w(t), [w^\star])$ vs t. Should $\to 0$.
- **Panel B (BR):** Same metric. Should $\to$ positive constant.
- **Panel C (Theorem 5 counterexample):** On game u with BR, plot $d(w(t), [w_u^\star])$ and $d(w(t), [w_v^\star])$. Both small (indistinguishable). With QR, only $d(w(t), [w_u^\star]) \to 0$.
- Games: Shapley $(3,3)$, $(2,)^{\times 5}$, counterexample $(4,2)$. $T = 5{,}000$. Seeds: 30.

3.3 Figure 3: Moderator vs No-Moderator — Swap Regret

Goal: Moderator gives $\mathcal{O}(\log T)$ swap regret, MW gives $\mathcal{O}(\sqrt{T})$.

- **Log-log plot.** x-axis: t (log). y-axis: avg player swap regret (log).
- **Curves:** Moderator (γ-BR), MW baseline, Regret Matching baseline.
- **Reference slopes:** $c_1 \log(t)$ and $c_2\sqrt{t}$.
- Games: Shapley, $(2,)^{\times 5}$, $(3,3,3)$. $T = 10{,}000$. Seeds: 30.
- $\gamma = 0.01$ for the moderator.

3.4 Figure 4: ε-CE Gap Convergence

Goal: Show the recommended distribution converges to a CE of the true game.

- x-axis: t. y-axis: ε-CE gap of $x(t)$ w.r.t. true utilities.
- **Curves:** Moderator (cutting-plane), Regret Matching, Random.
- Games: Shapley, Prisoner's Dilemma, $(2,)^{\times 5}$. $T = 5{,}000$. Seeds: 30.
- **Expected:** Moderator $\to 0$ fast (once CE learned, gap is zero). Regret Matching $\to 0$ at $\mathcal{O}(1/\sqrt{T})$. Random stays high.

4 Work Plan: Phase 3 — Appendix Experiments (6 Figures/Tables)

Estimated time: 4–5 hours running.

4.1 Figure A1: Regret Scaling with nM (Theorem 8)

- Three curves varying n (fixed m=2), m (fixed n=2), m (fixed n=3).
- x-axis: nM (log). y-axis: regret at T=2,000 (log). Reference: slope-1 line.
- Seeds: 30.

4.2 Figure A2: Number of Deviations vs T and nM

- Panel A: Cumulative deviations vs t (semi-log). Panel B: Final deviations vs nM.
- Seeds: 30.

4.3 Figure A3: Effect of γ on Swap Regret

- $\gamma \in \{0.001, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$.
- Panel A: Final swap regret vs γ. Panel B: Final compliance regret vs γ.
- Reference: $(1 + 1/\gamma) \cdot nM \cdot \log T$.
- Games: Shapley, $(2,)^{\times 5}$, $(3,3,3)$. $T = 5{,}000$. Seeds: 20.

4.4 Figure A4: Platform vs Player Regret (Joint View)

- Three curves: platform compliance, avg swap, max swap.
- Games: Shapley, $(2,)^{\times 5}$, $(3,3,3)$. $T = 5{,}000$. Seeds: 20.

4.5 Figure A5: Wall-Clock Time Scaling

- Panel A: Time per round vs nM. Breakdown: CE solve, ACCPM, oracle.
- Panel B: Time per round vs number of accumulated cuts.
- Games: $(2,2)$ through $(2,)^{\times 12}$. $T = 200$. Seeds: 3.

4.6 Table A1: Maximum Feasible Scale

- Game, n, M, nM, N, avg time/round, deviations, final regret, memory.
- Games: $(2,)^{\times 10}$, $(2,)^{\times 12}$, $(10, 10, 10)$, $(50, 50)$.
- Seeds: 3. $T = 500$.

5 Work Plan: Phase 4 — Compact Pipeline for Congestion Games

Estimated time: 1–2 days coding + testing.

5.1 Task 4.1: Implement congestion game environment

- Network $G = (V, E)$ with source-destination pairs.
- Affine edge costs: $c_e(\ell) = \theta_{e,1} + \theta_{e,2}\ell$.
- Route enumeration (all simple s_i–t_i paths up to length bound).
- Feature vectors $\phi_i(a)$ and difference features $\Delta\phi_i(r, r', a_{-i})$.
- Parameter dimension $d = 2|E|$.

5.2 Task 4.2: Implement compact CE solver

- Product-distribution CCE: variables $f_{i,r}$ (marginals per agent per route).
- LP with $\sum_i m_i$ variables instead of $M = \prod_i m_i$.
- Incentive constraints using expected congestion moments.

5.3 Task 4.3: Implement compact oracle

- Cut directly in $\mathbb{R}^d$: $q_\theta = \Phi^\top q$ computed via expected congestion.
- No enumeration of opponent profiles.
- BR/QR feedback using marginals.

5.4 Task 4.4: Implement compact cutting-plane

- Reuse ACCPM in $\mathbb{R}^d$ (already implemented, just change dimension).
- Per-round: $\theta(t) \to$ reconstruct costs $\to$ compact CE $\to$ sample $\to$ observe $\to$ cut in $\mathbb{R}^d$.

5.5 Task 4.5: Congestion game experiments

Figure 5 (Main paper): Compact pipeline on congestion games.

- **Panel A:** Braess network (4 nodes, 5 edges, $n = 10$–100 agents, $d = 10$). Regret vs T.
- **Panel B:** Sioux Falls network (24 nodes, 76 edges, $n = 100$–500, $d = 152$). Regret vs T.
- **Panel C:** Scaling: regret at T=2,000 vs number of agents n (fixed network).
- Baselines: myopic routing (current estimate only), oracle, random.
- $T = 2{,}000$. Seeds: 30.
- Compare runtime: compact vs normal-form (on Braess where both are feasible).

Figure A6 (Appendix): Parameter recovery in congestion game.

- $\|\theta(t) - \theta^\star\|/\|\theta^\star\|$ vs t on Braess network.
- BR vs QR comparison (Theorems 4–5 in the congestion setting).

6 Work Plan: Phase 5 — Final Polish

Estimated time: 1 day.

1. **Plot formatting:** NeurIPS template sizing (single/double column), 8–9pt fonts, colorblind palette, PDF export with tight bounding boxes.
2. **Error bands:** All plots with shaded 20th–80th percentile bands.
3. **Statistical summary table:** For each experiment, report median ± IQR of final metrics.
4. **Reproducibility:** Fixed seed sequences, requirements.txt, README with instructions.
5. **Commit and tag:** `git tag v1.0-neurips-submission`.

7 Timeline

Phase	**Content**	**Est. Time**
1	Benchmarks, metrics, baselines	3 hours
2	Main paper figures (4)	4 hours
3	Appendix figures (6)	5 hours
4	Compact congestion pipeline	12 hours
5	Polish and reproducibility	4 hours
	Total	**28 hours**

8 Summary: Figures for the Paper

#	Figure	Theorem	Key message	Location
1	Regret vs T	Thm 8	$\mathcal{O}(nM \log T)$ compliance regret	Main
2	QR vs BR convergence	Thm 4+5	QR learnable, BR not	Main
3	Moderator vs MW	Swap regret	$\mathcal{O}(\log T)$ vs $\mathcal{O}(\sqrt{T})$	Main
4	ε-CE gap	Thm 8	Convergence to CE	Main
5	Congestion games	Compact	Scales to n=500 agents	Main
A1	Regret vs nM	Thm 8	Linear in nM	Appendix
A2	Deviations vs T	Thm 8	Learning efficiency	Appendix
A3	Effect of γ	Swap regret	Tradeoff curve	Appendix
A4	Platform vs player	Joint	Both controlled	Appendix
A5	Wall-clock time	Feasibility	Runtime breakdown	Appendix
A6	Congestion recovery	Thm 4+5	θ-convergence	Appendix
T1	Max scale table	Scalability	Limits of normal-form	Appendix